KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)

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SDC1-WEEK2-DAY3

II-I CSE, CSE(AI&ML)

AY: 2026-2027

SL No	Week	Topic	Detailed Subtopics
1	Week2	KNN	KNN Introduction, Applications, Working Principle, Significance and Selection of K, Distance Metrics, Feature Scaling, Training and Prediction Phases, Advantages and Limitations

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(Week 2-Session 1)

1. What is KNN?
2. Where is KNN Used
3. Working Principle
4. What is “K” in KNN?
5. How to choose the value of K in KNN?
6. Distance Metrics used in KNN
7. Why Feature Scaling matters
8. Training Phase vs Prediction phase
9. Advantages
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1.What is KNN?

K-Nearest Neighbours (KNN) is a supervised, non-parametric, instance-based machine-learning algorithm. It predicts the output for a new observation by examining the K most similar observations in the labelled training data.

- Supervised: training examples contain known target labels or values.
- Non-parametric: it does not assume a fixed equation or probability distribution for the data.
- Instance-based / lazy learner: it stores training examples and performs most computation when a prediction is requested.

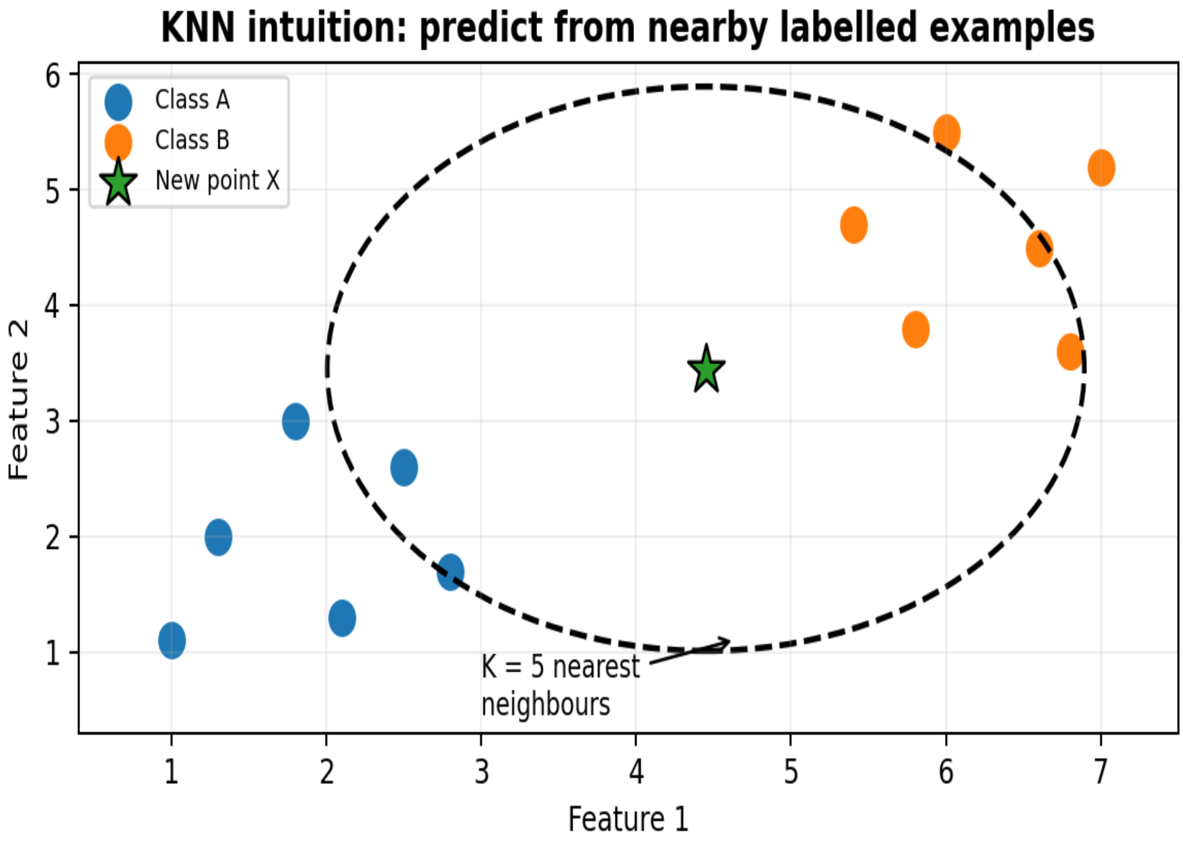


Figure 1. A new point is classified using the labels of its nearest neighbours.

Core intuition

KNN follows a simple idea: observations that are close to one another in feature space are often similar. If most nearby points belong to Class A, the new point is likely to belong to Class A as well.

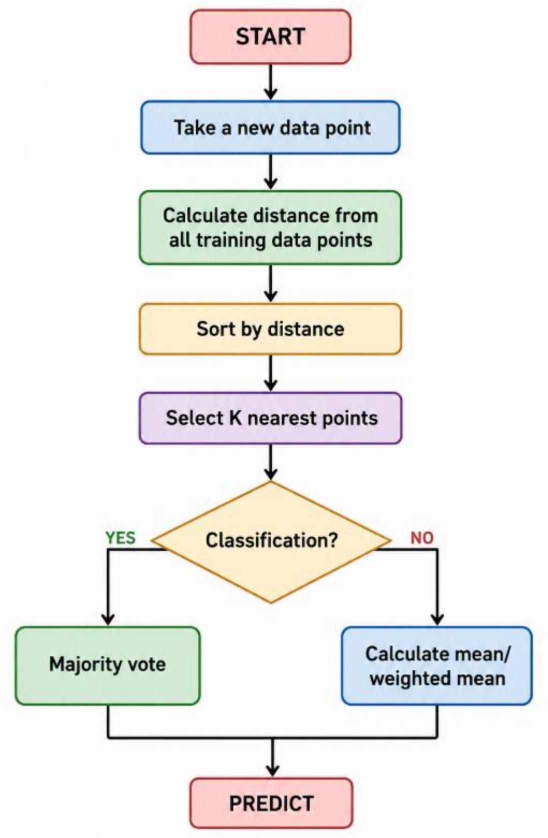
New point → Measure distance → Find K nearest → Vote / Average → Predict

2.Where is KNN used?

Problem	Example input features	Typical output
Classification	Age, income, spending score	Customer segment
Classification	Symptoms, test values	Disease category
Regression	Area, rooms, locality score	House price
Pattern matching	Image or signal features	Most similar class

Student memory line: K tells us how many neighbours are allowed to influence the prediction.

3.Working Principle



Algorithm steps

1. Choose a value for K (for example, K = 3 or K = 5).
2. Take the new observation whose output must be predicted.
3. Calculate its distance from each training observation.
4. Sort the training observations from smallest distance to largest distance.
5. Select the K closest observations.
6. For classification, use majority voting; for regression, use the mean or weighted mean.
7. Return the resulting class label or numerical prediction.

4. What is 'K' in K Nearest Neighbour?

In the k-Nearest Neighbours algorithm k is just a number that tells the algorithm how many nearby points or neighbors to look at when it makes a decision.
Example: Imagine you're deciding which fruit it is based on its shape and size. You compare it to fruits you already know.

- If k = 3, the algorithm looks at the 3 closest fruits to the new one.
- If 2 of those 3 fruits are apples and 1 is a banana, the algorithm says the new fruit is an apple because most of its neighbors are apples.

5. How to choose the value of k for KNN Algorithm?

- The value of k in KNN decides how many neighbors the algorithm looks at when making a prediction.
- Choosing the right k is important for good results.

- If the data has lots of noise or outliers, using a larger k can make the predictions more stable.
- But if k is too large the model may become too simple and miss important patterns and this is called underfitting.
- So k should be picked carefully based on the data.

6. Distance Metrics Used in KNN Algorithm

KNN uses distance metrics to identify nearest neighbor, these neighbors are used for classification and regression task. To identify nearest neighbor we use below distance metrics:

a) Euclidean Distance

Euclidean distance is defined as the straight-line distance between two points in a plane or space. You can think of it like the shortest path you would walk if you were to go directly from one point to another.
For two points $A(x_1, y_1)$ and $B(x_2, y_2)$, Euclidean distance is the ordinary straight-line distance:

$$d(A,B) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

Example: A = (2, 3), B = (5, 7). Therefore $d = \sqrt{(3^2 + 4^2)} = 5$.

b) Manhattan Distance

This is the total distance you would travel if you could only move along horizontal and vertical lines like a grid or city streets. It’s also called "taxicab distance" because a taxi can only drive along the grid-like streets of a city.

$$d(A,B) = =|x2-x1|+|y2-y1|$$

For A = (2, 3) and B = (5, 7), Manhattan distance = 3 + 4 = 7. It is useful when movement or difference is naturally measured along dimensions rather than as a straight line.

c) Minkowski Distance

Minkowski distance is like a family of distances, which includes both Euclidean and Manhattan distances as special cases.

$$D(x, y) = \left(\sum |xi - yi|^p\right)^{\frac{1}{p}}$$

From the formula above, when p=2, it becomes the same as the Euclidean distance formula and when p=1, it turns into the Manhattan distance formula. Minkowski distance is essentially a flexible formula that can represent either Euclidean or Manhattan distance depending on the value of p.

7. Why feature scaling matters

Suppose a dataset contains Age (18–70) and Annual Salary (₹2,00,000–₹50,00,000). A salary difference of 500,000 can dominate an age difference of 5, even if age is equally important. Because KNN is distance-based, differently scaled features can distort the neighbourhood.

Standardization: $z = (x - \mu) / \sigma$

Min–max normalization: $x' = (x - xmin) / (xmax - xmin)$

Rule : split the data first; fit the scaler on the training set only; then use that fitted scaler to transform validation/test data. This avoids data leakage.

7. Training Phase Vs Prediction Phase

Phase	What KNN Does
Training	Stores the training data
Prediction	Calculates distance to training points
Neighbour search	Finds K closest samples
Classification	Takes majority vote
Regression	Calculates average

Unlike algorithms such as linear regression or decision trees, KNN does not construct a conventional mathematical model during training.

8. Advantages of KNN

KNN is popular because it is:

- Simple to understand.
- Easy to implement.
- Useful for both classification and regression.
- Non-parametric.
- Able to model nonlinear decision boundaries.
- Naturally adaptable to multi-class classification.
- Effective for small and moderately sized datasets when the features are meaningful.

9. Limitations of KNN

KNN also has important disadvantages:

- Prediction can be slow for large datasets.
- It requires storing the training data.
- It is sensitive to feature scaling.
- It can be affected by noisy data and outliers.
- Choosing K is important.
- Irrelevant features can reduce performance.
- Performance may deteriorate when there are many dimensions.

K-Nearest Neighbors (KNN) – Solved Numerical Example Problem

Problem

A college wants to classify a student as Normal or Underweight based on Height and Weight. The following training data is available. A new student has a height of 170 cm and weight of 57 kg. Using K = 3, predict the class of the new student.

Student	Height (cm)	Weight (kg)	Class
A	169	58	Normal
B	170	55	Normal
C	173	57	Normal
D	174	56	Underweight
E	167	51	Underweight
F	173	64	Normal
G	172	65	Normal
H	182	62	Normal
I	176	69	Normal

Step 1: Identify the Query Point

The new student is the query point:

$X_q = (170, 57)$

Step 2: Choose K

The problem specifies K = 3. Therefore, we must find the 3 training students closest to the new student.

Step 3: Calculate Euclidean Distance

For two points, Euclidean distance is calculated as:

$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Example: Distance between the query point (170, 57) and Student A (169, 58):

$d = \sqrt{[(169 - 170)^2 + (58 - 57)^2]}$

$= \sqrt{[(-1)^2 + (1)^2]}$

$= \sqrt{2}$

$= 1.414$

Step 4: Calculate Distance for Every Student

Student	Class	Distance
A	Normal	1.414
B	Normal	2.000
C	Normal	3.000
D	Underweight	4.123
E	Underweight	6.708
G	Normal	8.246
F	Normal	8.602
I	Normal	13.454
H	Normal	13.601

Step 5: Rank the Distances

Arrange the distances from smallest to largest. The smallest distance represents the closest neighbor.

Rank	Student	Distance	Class
1	A	1.414	Normal
2	B	2.000	Normal
3	C	3.000	Normal
4	D	4.123	Underweight
5	E	6.708	Underweight

Step 6: Select the K Nearest Neighbors

Since K = 3, select the first three students:

A → Normal

B → Normal

C → Normal

Step 7: Majority Voting

For classification, KNN uses majority voting. Among the 3 nearest neighbors:

Normal = 3 votes

Underweight = 0 votes

Final Prediction

The new student is classified as NORMAL.

KNN Procedure to Remember

1.Calculate the distance between the query point and every training point.

2.Arrange the distances from smallest to largest.
3.Select the K nearest neighbors.
4.For classification, count the class votes.
5.The class with the highest number of votes becomes the prediction.

Q & A

Q1. What is KNN?
Answer: KNN is a supervised machine learning algorithm used for classification and regression.
It predicts a new data point based on its **K nearest neighbors**.

Q2. What does K represent in KNN?
Answer: K represents the number of nearest neighbors considered for prediction.
For example, means the algorithm considers the **5 closest data points**.

Q3. A KNN model is trained on a dataset containing 100 training samples. The developer chooses . What is likely to happen to the model's prediction, and why?
Answer: The model will consider almost the entire training dataset for every prediction, so local patterns will be largely ignored.
This can make the model too generalized, resulting in high bias and possible underfitting.

Q4. Why is feature scaling important in KNN?
Answer: KNN is distance-based, so features with larger numerical ranges can dominate the distance calculation.
Scaling ensures that all features contribute more fairly to the distance.

B. Scenario-Based Questions – 6
Q5. Student Result Prediction
A KNN model uses . The five nearest students have results: **Pass, Pass, Fail, Pass, Fail**. What will be the prediction?
Answer: Pass.
Pass receives 3 votes while Fail receives 2 votes, so KNN predicts the majority class, **Pass**.

Q6. Customer Classification
A company uses Age and Income to classify customers. Age ranges from 18–70, while Income ranges from ₹20,000–₹20,00,000. What should be done before applying KNN?
Answer: Feature scaling should be performed.
Income has a much larger range and may dominate the distance calculation without scaling.

Q7. A KNN model is tested with two different values of . With , the model gives 98% training accuracy but 70% testing accuracy. With , it gives 75% training accuracy and 74% testing accuracy. What do these results indicate?
Answer: may cause overfitting because the model is highly sensitive to individual training points, while may cause underfitting because it considers too many neighbors.
Therefore, an appropriate moderate value of K should be selected to achieve better generalization.

Q8. House Price Prediction
The 3 nearest houses have prices of **₹40 lakh, ₹45 lakh, and ₹50 lakh**. What is the predicted price using KNN regression with ?
Answer: ₹45 lakh.
KNN regression takes the average: lakh.

Q9. Selecting the Best K

A student obtains the following test accuracies: → 82%, → 88%, → 93%, → 89%. Which K should be selected?

Answer: .

It gives the highest testing accuracy of **93%**, so it is the best choice among the tested values.

Q10. Medical Diagnosis

A hospital applies KNN using Age, Blood Pressure, and Cholesterol. The model performs poorly because cholesterol values are much larger numerically than age. What should be done?

Answer: Scale the features before applying KNN.

Since KNN uses distance, scaling prevents the larger-valued cholesterol feature from dominating the distance calculation.